

Ch 10: The Human Eye and the Colourful World

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is meant by the power of accommodation of the eye? Why is it important? (PYQ 2019)

Q2. Name the light-sensitive screen of the human eye. What type of image is formed on it? (PYQ 2020)

Q3. What is the function of the eye lens? Why should the eye lens be flexible? (PYQ 2022)

Q4. Define near point and far point of the normal human eye. Write their usual values. (PYQ 2019)

Q5. What is the least distance of distinct vision for a normal eye? State its value. (PYQ 2020)

Q6. What is myopia? What type of lens is used to correct it? (PYQ 2022)

Q7. What is hypermetropia? What type of lens is used to correct it? (PYQ 2019)

Q8. What is presbyopia? Why does it usually occur in old age? (PYQ 2020)

Q9. What is cataract? How is vision restored in a cataract patient? (PYQ 2022)

Q10. What is meant by dispersion of white light? Name the optical device used to show dispersion. (PYQ 2019)

Q11. Name the seven colours obtained when white light passes through a glass prism. Write them in order. (PYQ 2020)

Q12. Which colour deviates the most and which colour deviates the least in dispersion through a prism? (PYQ 2022)

Q13. What is spectrum? How is the spectrum of white light produced by a prism? (PYQ 2019)

Q14. What is recombination of spectrum? Name the device used to recombine the colours of white light. (PYQ 2020)

Q15. What is atmospheric refraction? Give one example of a phenomenon caused by it. (PYQ 2022)

Q16. Why do stars twinkle but planets do not twinkle significantly? (PYQ 2019)

Q17. Why does the sun appear slightly higher than its actual position during sunrise and sunset? (PYQ 2020)

Q18. What is the advance sunrise and delayed sunset? How much extra daylight do they approximately give? (PYQ 2022)

Q19. What is scattering of light? Name the effect that explains scattering by fine particles. (PYQ 2019)

Q20. What is Tyndall effect? Give two examples of Tyndall effect from daily life. (PYQ 2020)

Q21. Why is the colour of the clear sky blue? (PYQ 2022)

Q22. Why does the sun appear reddish at sunrise and sunset? (PYQ 2019)

Q23. Why are danger signals red in colour? (PYQ 2020)

Q24. Why does the sky appear dark instead of blue to astronauts in space? (PYQ 2022)

Q25. What is a rainbow? Which natural phenomena are involved in its formation? (PYQ 2019)

Q26. Why does a rainbow form after rainfall when sunlight is present? (PYQ 2020)

Q27. Name the part of the eye that controls the amount of light entering the eye. (PYQ 2022)

Q28. What is the function of the pupil in the human eye? (PYQ 2019)

Q29. What is the function of the ciliary muscles in the eye? (PYQ 2020)

Q30. What is the blind spot in the human eye? Why is no image formed at this spot? (PYQ 2022)

Q31. What is the role of the optic nerve in vision? (PYQ 2019)

Q32. Name the transparent front part of the eye. What is its function? (PYQ 2020)

Q33. What is meant by persistence of vision? State one application of this phenomenon. (PYQ 2022)

Q34. Why do we need two eyes for vision? State any two advantages. (PYQ 2019)

Q35. What is stereoscopic vision? How does it help human beings? (PYQ 2020)

Q36. What happens to the eye lens when we look at a nearby object? (PYQ 2022)

Q37. What happens to the eye lens when we look at a distant object? (PYQ 2019)

Q38. Why is a convex lens used to correct hypermetropia? (PYQ 2020)

Q39. Why is a concave lens used to correct myopia? (PYQ 2022)

Q40. Differentiate between scattering and dispersion in one major point. (PYQ 2024)

SECTION B - 3 MARKS

Q1. Draw a labelled diagram of the human eye and label cornea, iris, pupil, eye lens, retina and optic nerve. State one function of each. (PYQ 2019)

Q2. Explain how the eye adjusts itself to see nearby and distant objects clearly. Include the role of ciliary muscles and eye lens. (PYQ 2020)

Q3. What is power of accommodation? Explain what happens to the focal length of the eye lens while seeing near and distant objects. (PYQ 2022)

Q4. Compare myopia and hypermetropia: (PYQ 2019)

Basis of Comparison	Myopia	Hypermetropia
Object not seen clearly		
Image formed		
Cause		
Lens used for correction		

Q5. Explain myopia with a ray diagram. Mention the cause of myopia and the method of correction. (PYQ 2020)

Q6. Explain hypermetropia with a ray diagram. Mention the cause of hypermetropia and the method of correction. (PYQ 2022)

Q7. What is presbyopia? Explain why bifocal lenses are used by some elderly people. (PYQ 2019)

Q8. Distinguish between cataract and presbyopia on the basis of cause and correction. (PYQ 2020)

Q9. Explain dispersion of white light by a glass prism with a labelled diagram. Why does violet deviate more than red? (PYQ 2022)

Q10. Describe Newton's experiment to show that sunlight is made up of seven colours. (PYQ 2019)

Q11. How can the seven colours of the spectrum be recombined to get white light? Explain with diagram. (PYQ 2020)

Q12. Explain the formation of a rainbow. Mention the role of refraction, dispersion and internal reflection. (PYQ 2022)

Q13. Explain atmospheric refraction. How does it cause twinkling of stars? (PYQ 2019)

Q14. Why do planets generally not twinkle whereas stars twinkle? Give a reason based on apparent size and atmospheric refraction. (PYQ 2020)

Q15. Explain why the sun is visible about two minutes before actual sunrise and about two minutes after actual sunset. (PYQ 2022)

Q16. What is Tyndall effect? Explain with three observations from everyday life. (PYQ 2019)

Q17. Explain why the sky is blue during the day but appears dark in space. (PYQ 2020)

Q18. Explain why the sun appears red at sunrise and sunset but white at noon. (PYQ 2022)

Q19. Why are danger signals and stop lights red? Explain using scattering of light. (PYQ 2019)

Q20. Compare dispersion, scattering and atmospheric refraction: (PYQ 2020)

Basis	Dispersion	Scattering	Atmospheric Refraction
Meaning			
Cause			
Example			

Q21. Explain the functions of the following parts of the eye: cornea, iris, pupil, retina, optic nerve and ciliary muscles. (PYQ 2022)

Q22. What is the near point of the eye? How does it change in hypermetropia and presbyopia? (PYQ 2019)

Q23. What is the far point of the eye? How does it change in myopia? How is the defect corrected? (PYQ 2020)

Q24. How do two eyes provide wider field of view and better depth perception? Explain. (PYQ 2022)

Q25. What is persistence of vision? Explain why motion pictures appear continuous to our eyes. (PYQ 2019)

Q26. What happens when: (i) ciliary muscles relax (ii) ciliary muscles contract (iii) pupil becomes small? (PYQ 2020)

Q27. Explain the path of light through the human eye from cornea to brain. (PYQ 2022)

Q28. Why should the image be formed exactly on the retina for clear vision? What happens if it forms in front of or behind the retina? (PYQ 2019)

Q29. Explain the role of rods and cones in the retina. What type of vision is each associated with? (PYQ 2020)

Q30. Draw ray diagrams for correction of myopia and hypermetropia and state the type of lens used in each case. (PYQ 2022)

Q31. Explain why white light splits into seven colours in a prism but not through a rectangular glass slab in the same way. (PYQ 2019)

Q32. Why does a beam of sunlight become visible in a dusty room? Which phenomenon is responsible? (PYQ 2020)

Q33. Explain the role of tiny water droplets in the formation of rainbow. Why is the observer's position important? (PYQ 2022)

Q34. Case based: A student cannot see the blackboard clearly but can read a book easily. Identify the defect, cause and correction. (PYQ 2024)

Q35. Case based: An elderly person uses spectacles with two different powers. Identify the defect and explain the use of bifocal lenses. (PYQ 2024)

SECTION C - 5 MARKS

Q1. Draw a neat labelled diagram of the human eye. Explain the structure and functions of cornea, iris, pupil, eye lens, retina, optic nerve and ciliary muscles. (PYQ 2022)

Q2. Explain the working of the human eye in detail. Describe how an image is formed on the retina and how the brain helps in vision. (PYQ 2020)

Q3. Describe the power of accommodation of the eye. Explain the changes in the eye lens while viewing nearby and distant objects. Add a labelled diagram. (PYQ 2019)

Q4. Explain the defect of myopia in detail. Include causes, position of image, ray diagram, correction and type of lens used. (PYQ 2022)

Q5. Explain the defect of hypermetropia in detail. Include causes, position of image, ray diagram, correction and type of lens used. (PYQ 2020)

Q6. Compare myopia, hypermetropia and presbyopia across five parameters: cause, image position, near/far vision, lens used and common age group. (PYQ 2019)

Q7. Draw ray diagrams to show correction of myopia and hypermetropia. Explain why concave and convex lenses are used respectively. (PYQ 2022)

Q8. Write a detailed note on cataract and presbyopia. Explain how each affects vision and how each is corrected. (PYQ 2020)

Q9. Describe dispersion of white light through a glass prism. Draw the labelled diagram, write the order of colours and explain why different colours deviate differently. (PYQ 2019)

Q10. Describe Newton's prism experiment. How did Newton prove that white light is made up of seven colours and that a prism does not create colours? (PYQ 2022)

Q11. Explain recombination of the spectrum. Draw a diagram showing two prisms recombining the colours of white light. (PYQ 2020)

Q12. Explain the formation of rainbow in detail. Include refraction, dispersion, internal reflection and emergence of light from raindrops. (PYQ 2019)

Q13. Describe atmospheric refraction. Explain twinkling of stars, apparent position of stars and non-twinkling of planets using atmospheric refraction. (PYQ 2022)

Q14. Explain advance sunrise and delayed sunset with the help of atmospheric refraction. Why does the sun appear flattened near the horizon? (PYQ 2020)

Q15. Describe scattering of light. Explain Tyndall effect and give three examples where it is observed. (PYQ 2019)

Q16. Why is the sky blue? Why does it appear dark to astronauts? Explain both using scattering of light. (PYQ 2022)

Q17. Why does the sun appear reddish during sunrise and sunset? Explain the role of path length and scattering of shorter wavelengths. (PYQ 2020)

Q18. Explain why danger signals are red. Connect your answer with wavelength, scattering and visibility through fog or smoke. (PYQ 2019)

Q19. Compare dispersion, scattering and atmospheric refraction in detail with causes, examples and optical conditions. (PYQ 2022)

Q20. Case Study: A person can read a newspaper clearly but cannot see distant road signs clearly. Identify the defect. Explain the cause, ray diagram and correction. (PYQ 2024 case-based)

Q21. Case Study: A person can see distant objects clearly but cannot read a book held at 25 cm. Identify the defect. Explain the cause, ray diagram and correction. (PYQ 2024 case-based)

Q22. Case Study: A narrow beam of sunlight entering a dusty room becomes visible, but the same beam is not visible in clean air. Explain the phenomenon and conditions required. (PYQ 2023 case-based)

Q23. Case Study: During a rainy afternoon, a student observes a rainbow opposite the sun. Explain why the rainbow forms opposite the sun and why water droplets act like tiny prisms. (PYQ 2023 case-based)

Q24. Write a comprehensive note on the human eye covering image formation, accommodation, least distance of distinct vision, two common defects and their correction. (PYQ 2022 comprehensive)

Q25. Write a comprehensive note on colourful phenomena in nature covering dispersion, rainbow, atmospheric refraction, scattering, blue sky and red sunset. (PYQ 2022 comprehensive)

Q26. Answer the following in a table: (i) part of eye controlling light (ii) screen of eye (iii) nerve carrying signals (iv) defect corrected by concave lens (v) reason for blue sky. (PYQ 2024)